

Supplementary Material for Evaluations of Dynamic Gaussian Splats versus Point Clouds for Sparse Captures

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This document contains the supplemental material of the paper *Evaluations of Dynamic Gaussian Splats versus Point Clouds for Sparse Captures* (S. Croci et al.).

I. STIMULI FROM THE GAUSSIAN SPLAT VS POINT CLOUD EXPERIMENT

Figure 1 shows a sample frame from the stimuli used in the subjective experiment with the point cloud (PC) and Gaussian splat (GS) reconstructions on the left and right, respectively. The subfigure captions indicate the names of the captures used to generate the reconstructions. In the experiment, we also used the same stimuli with the left–right order of the reconstructions reversed.

II. ANALYSIS OF THE STIMULI WATCHING TIMES

This section presents the analysis of the stimuli watching times collected during the subjective experiment. In particular, Figure 2 shows the histogram of the overall watching times, while Figure 3 shows the per-capture histograms for the GS vs. PC and PC vs. GS stimuli. In these histograms, there are peaks close to 10 s, i.e., the duration of the stimuli. Moreover, Figure 4 shows a bar chart of the median watching time for the GS vs. PC and PC vs. GS stimuli of each capture, with error bars indicating the interquartile range (IQR). In this chart, the medians are close to 10 s.

III. ANALYSIS OF THE DYNAMIC GAUSSIAN SPLAT RECONSTRUCTIONS

This section presents the analysis of all dynamic GS reconstructions in Figure 5, and the analyses of each dynamic GS reconstruction in the Figures 6, 7, 8, 9, 10, and 11. As can be observed, the analyses of the dynamic GS reconstructions do not vary much.

IV. ANALYSIS OF THE STATIC GAUSSIAN SPLAT RECONSTRUCTIONS

This section presents the analyses of the static GS reconstructions of the paper [1] generated with a GS scene optimization solution called Postshot [2]. Figure 12 shows the analysis of all static GS reconstructions, while Figures

13, 14, 15, 16, 17, 18, and 19 show the analyses of each static GS reconstruction. It can be observed that there is a clear difference between the analyses of the dynamic GS reconstructions of our paper and the analyses of the static GS reconstructions.

REFERENCES

- [1] P. Haslbauer, M. Pullen, C. Reichherzer, and A. Smolic, “Gaussian splatting vs. classical photogrammetry: A comparison for virtual backdrops,” in *2025 17th International Conference on Quality of Multimedia Experience (QoMEX)*. IEEE, 2025, pp. 1–4.
- [2] (2025) Postshot. [Online]. Available: <https://www.jawset.com/>

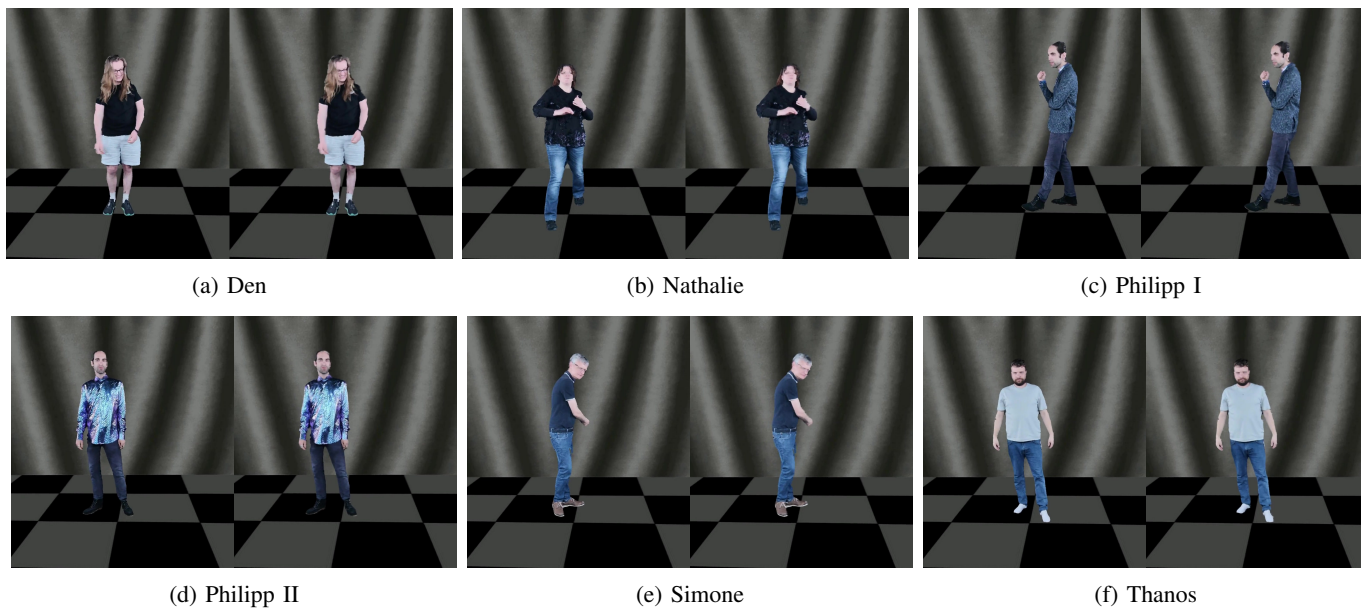


Fig. 1: Stimuli with point cloud and Gaussian splat reconstructions on the left and right, respectively, together with the names of the captures used to generate the reconstructions. In the subjective experiment, we also used the same stimuli with the reconstruction order reversed.

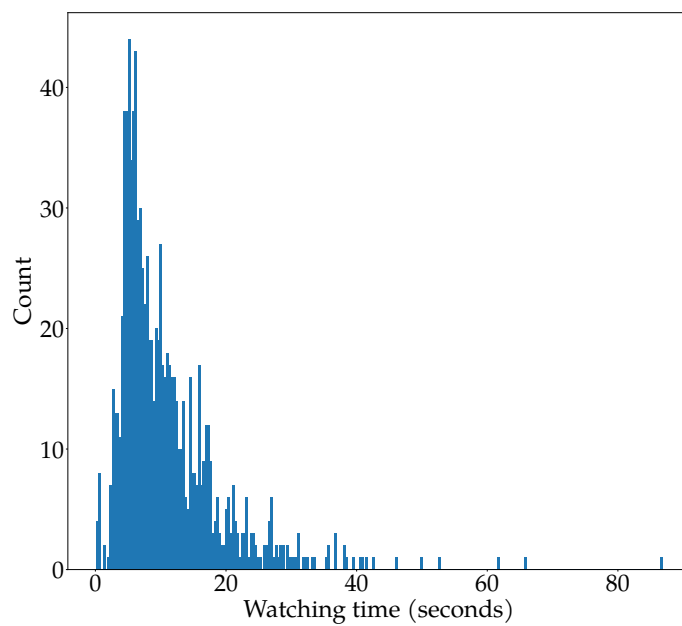


Fig. 2: Stimuli watching time histogram.

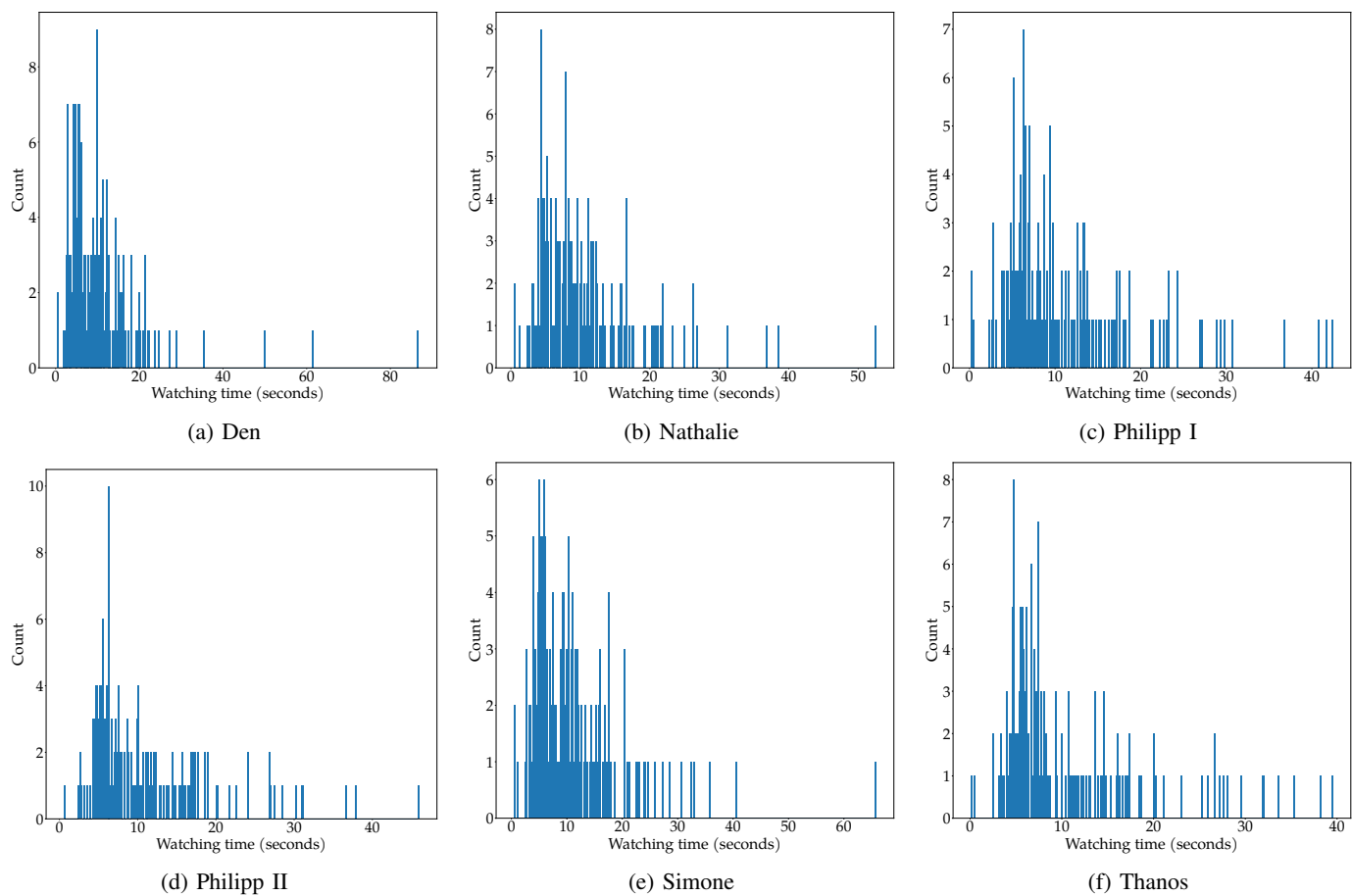


Fig. 3: Per-capture watching time histograms for the GS vs. PC and PC vs. GS stimuli.

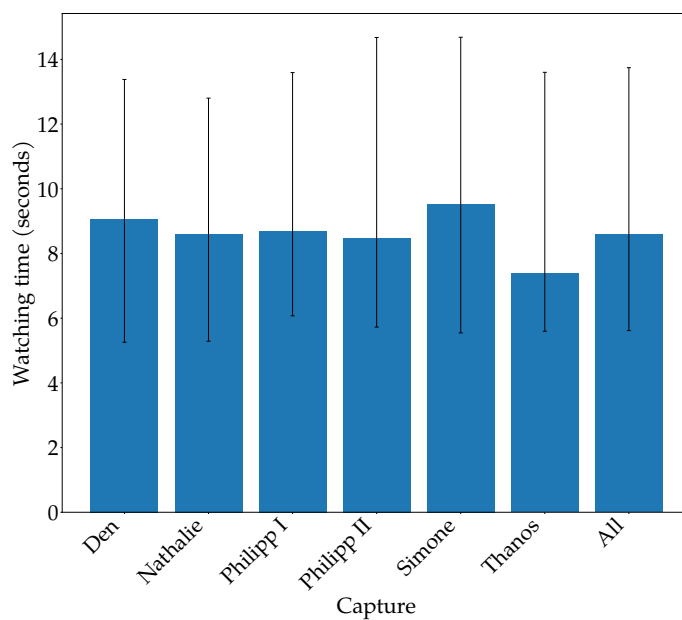


Fig. 4: Bar chart of the per-capture watching times for the GS vs. PC and PC vs. GS stimuli (bars = median, error bars = interquartile range).

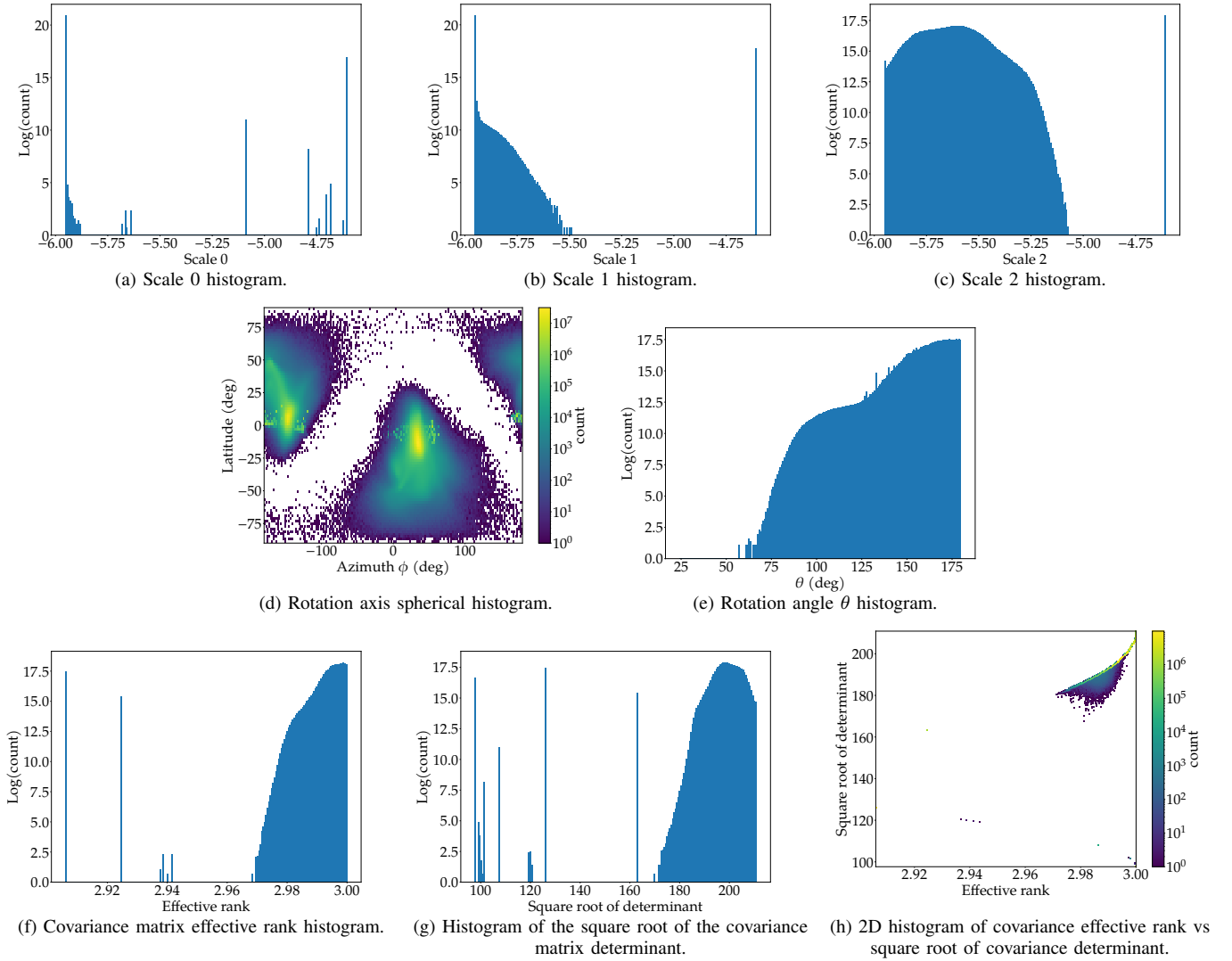


Fig. 5: Analysis of all dynamic Gaussian splat reconstructions.

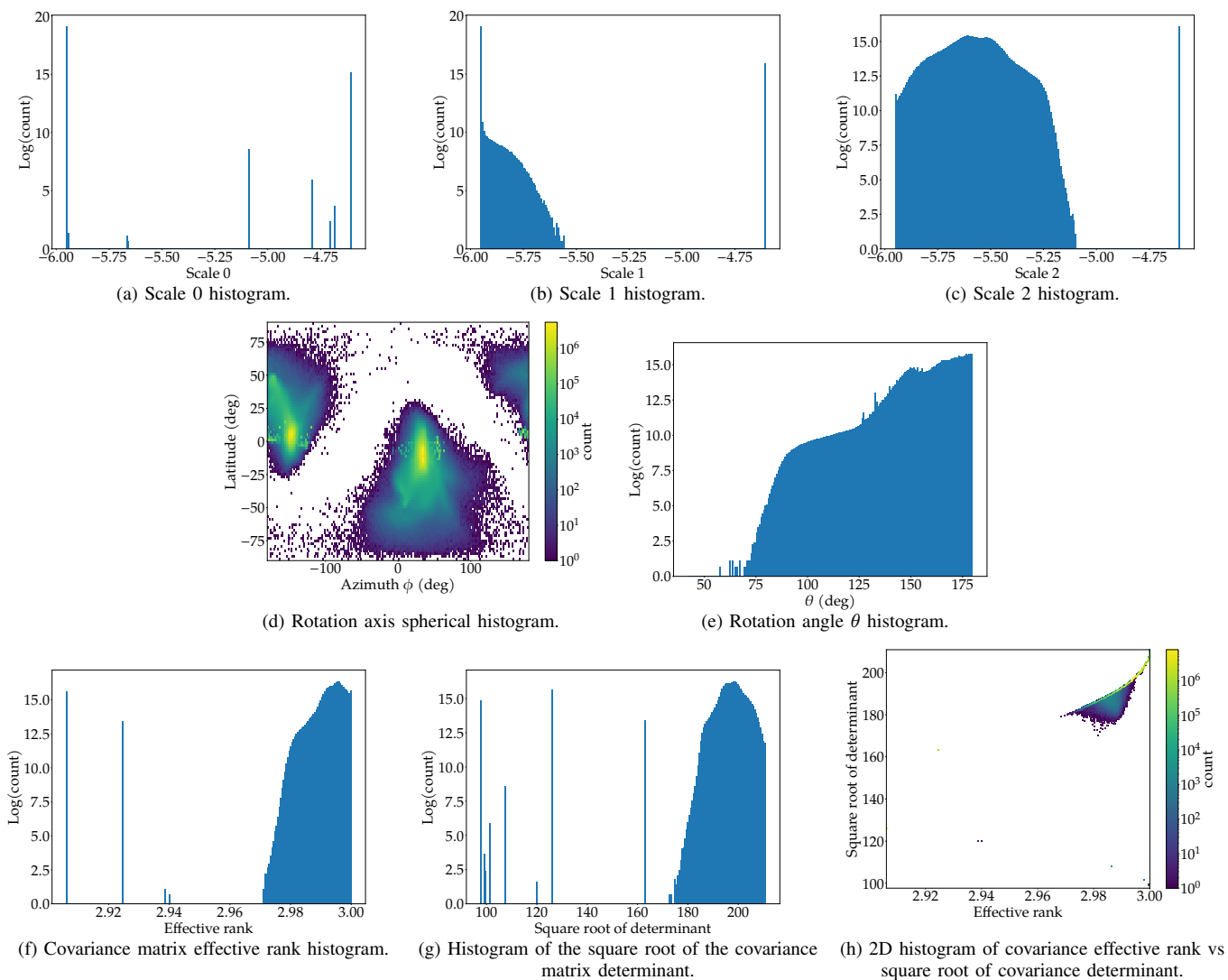


Fig. 6: Analysis of Den's dynamic Gaussian splat reconstruction.

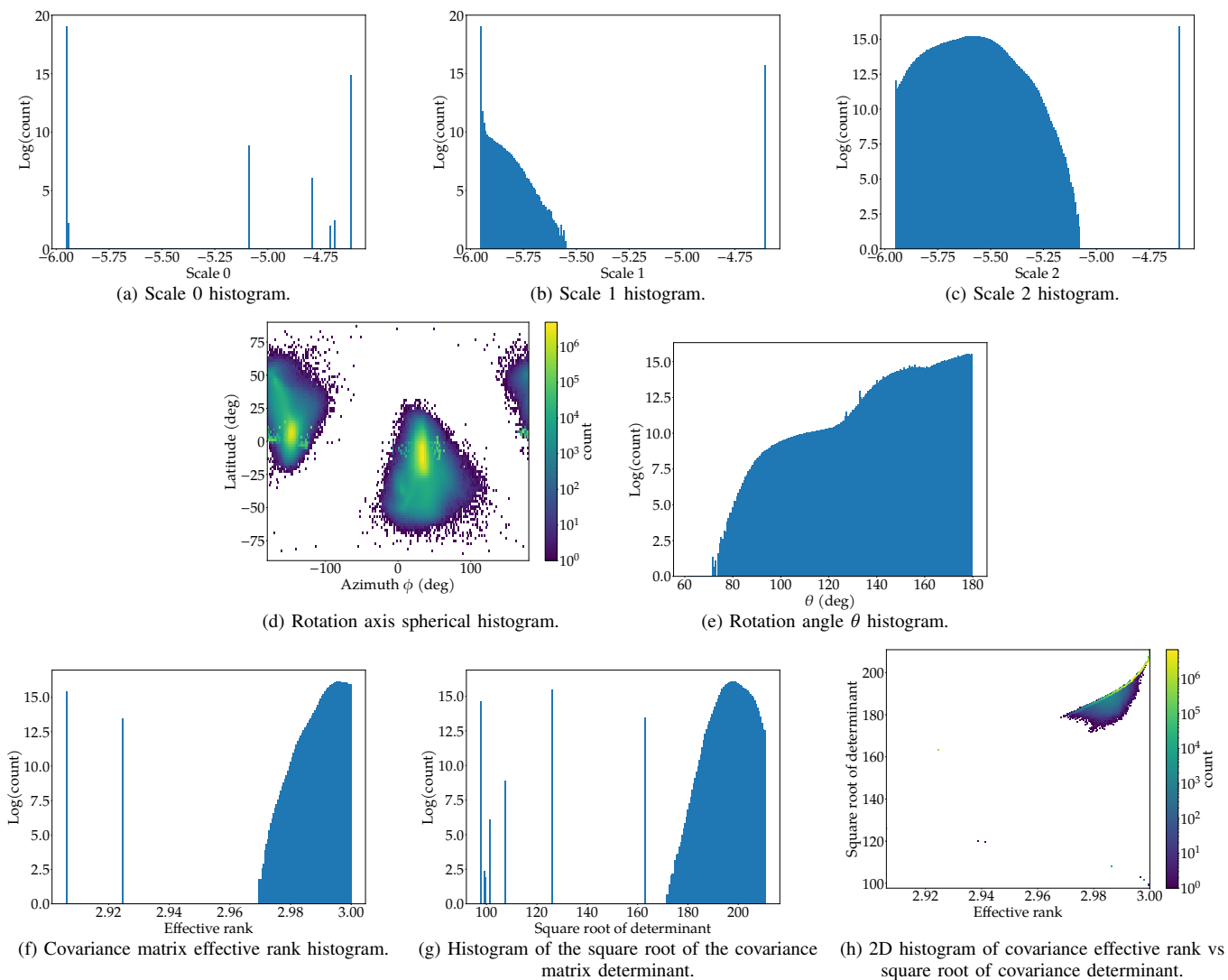


Fig. 7: Analysis of Nathalie's dynamic Gaussian splat reconstruction.

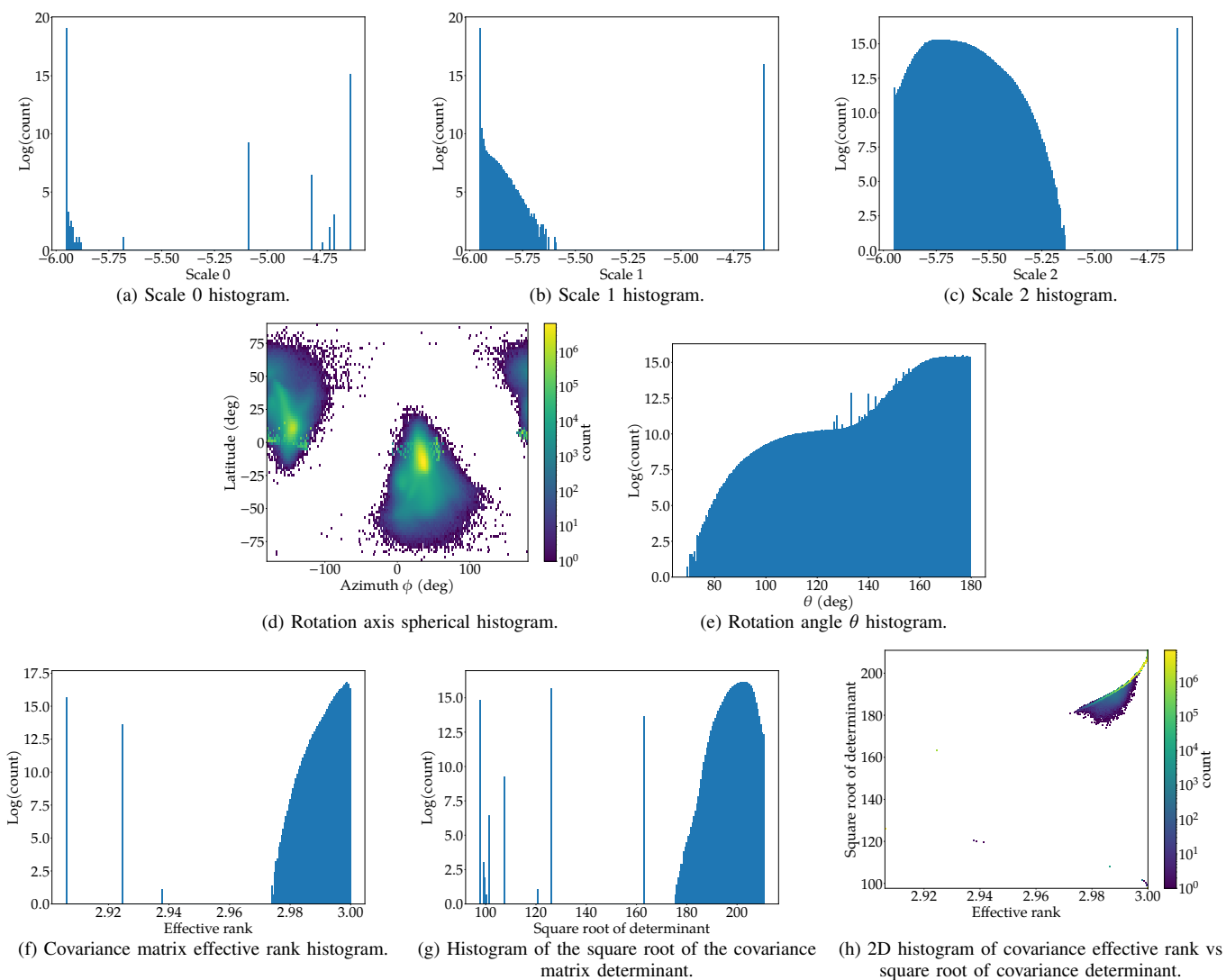


Fig. 8: Analysis of Philipp I's dynamic Gaussian splat reconstruction.

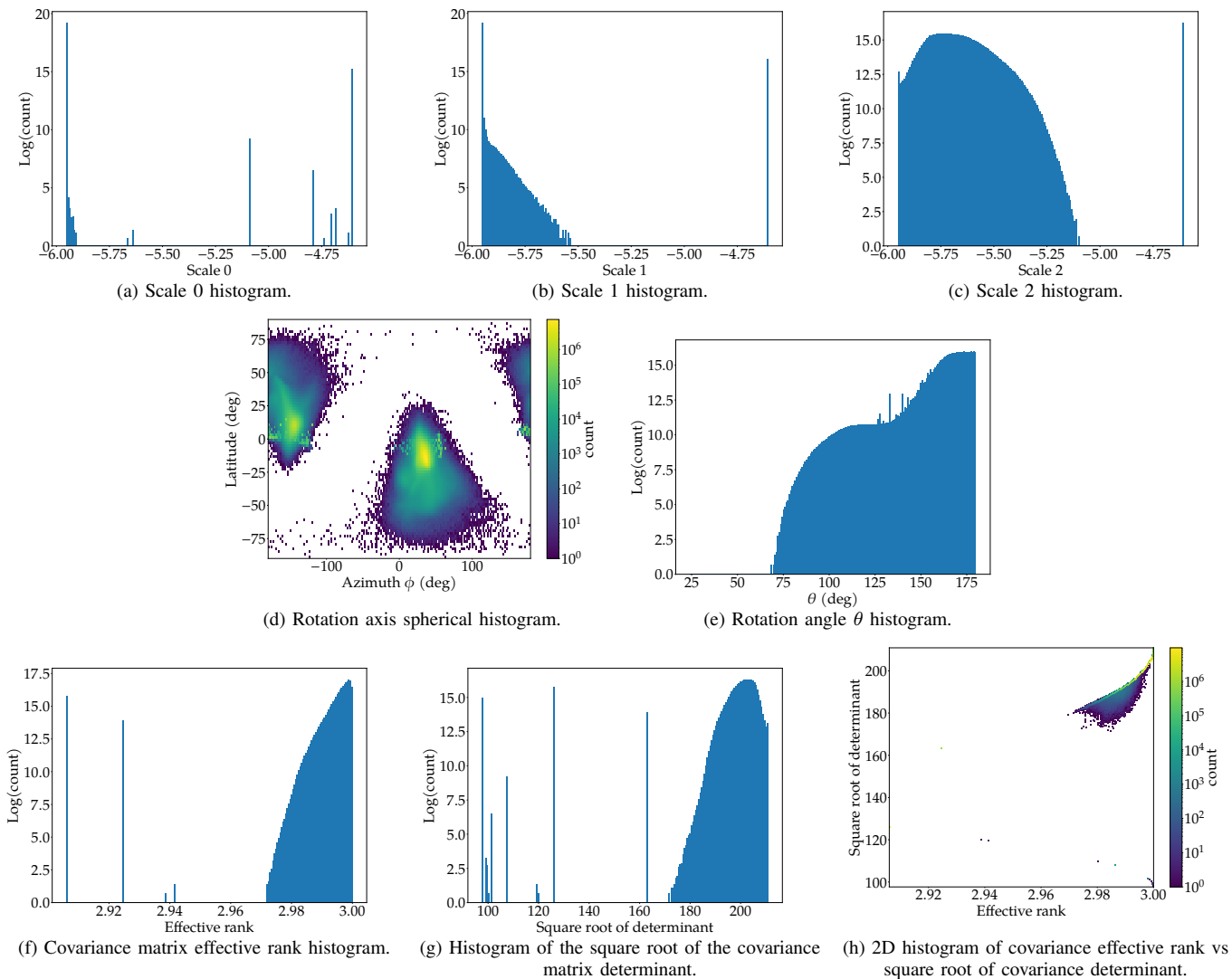


Fig. 9: Analysis of Philipp II's dynamic Gaussian splat reconstruction.

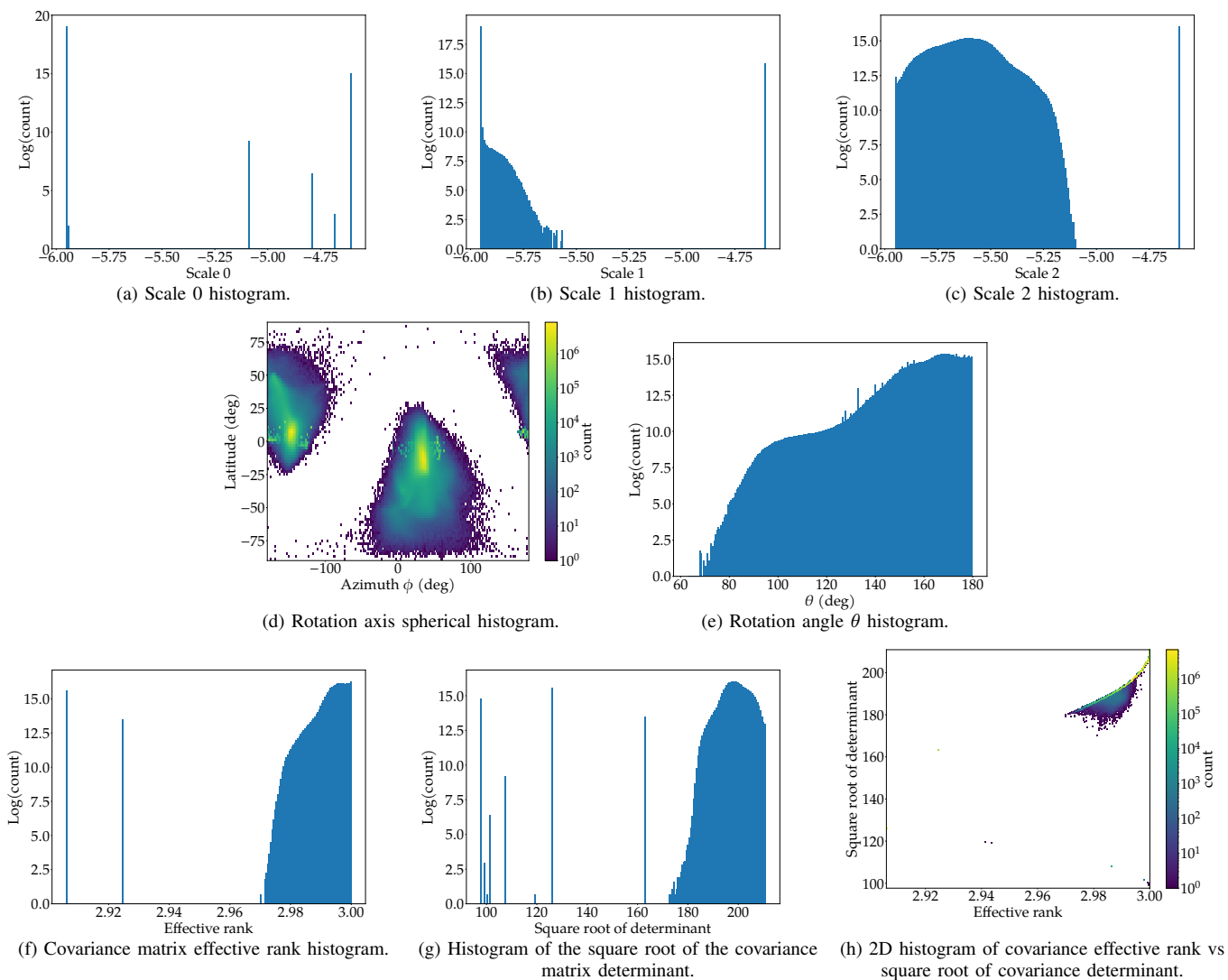


Fig. 10: Analysis of Simone's dynamic Gaussian splat reconstruction.

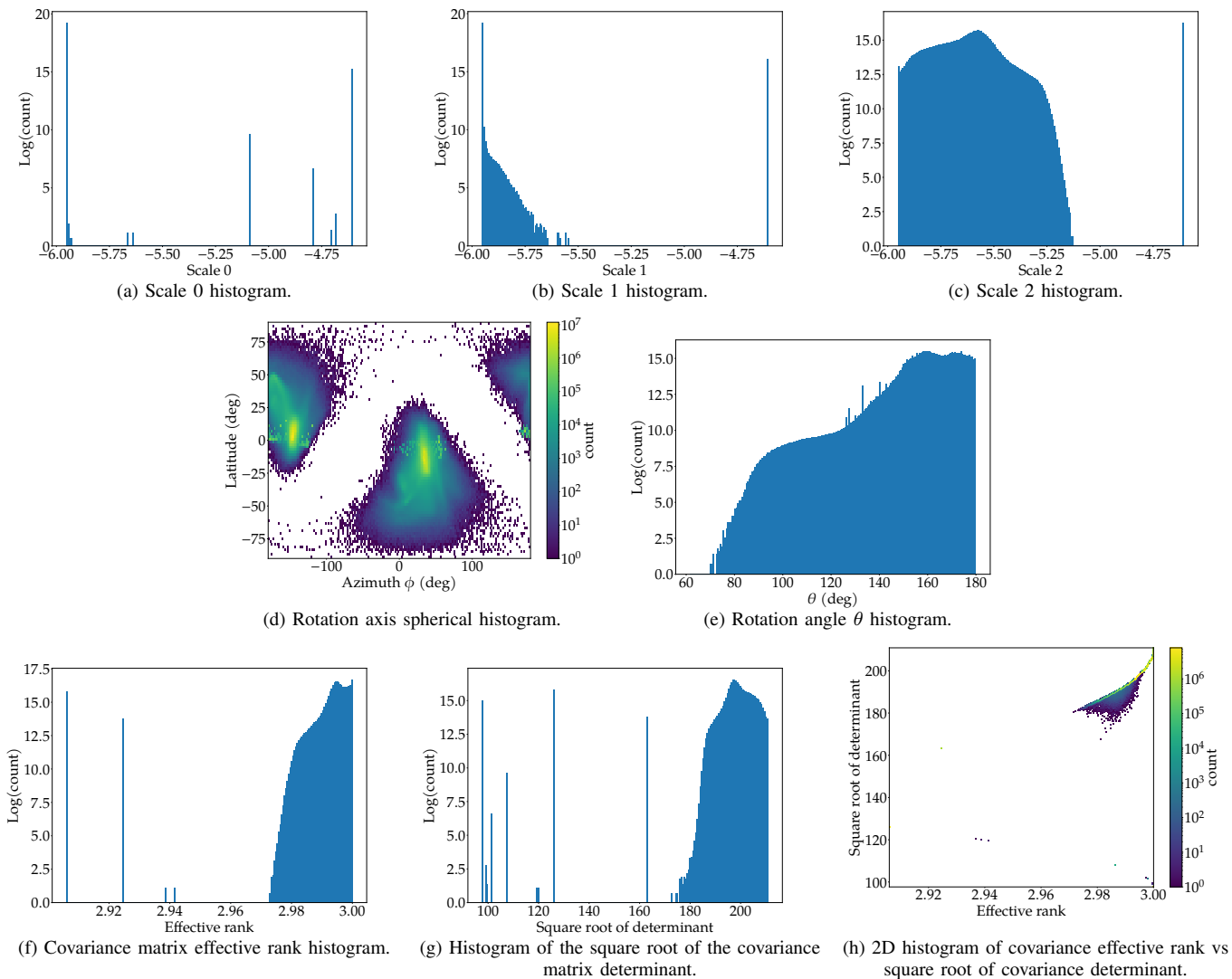


Fig. 11: Analysis of Thanos's dynamic Gaussian splat reconstruction.

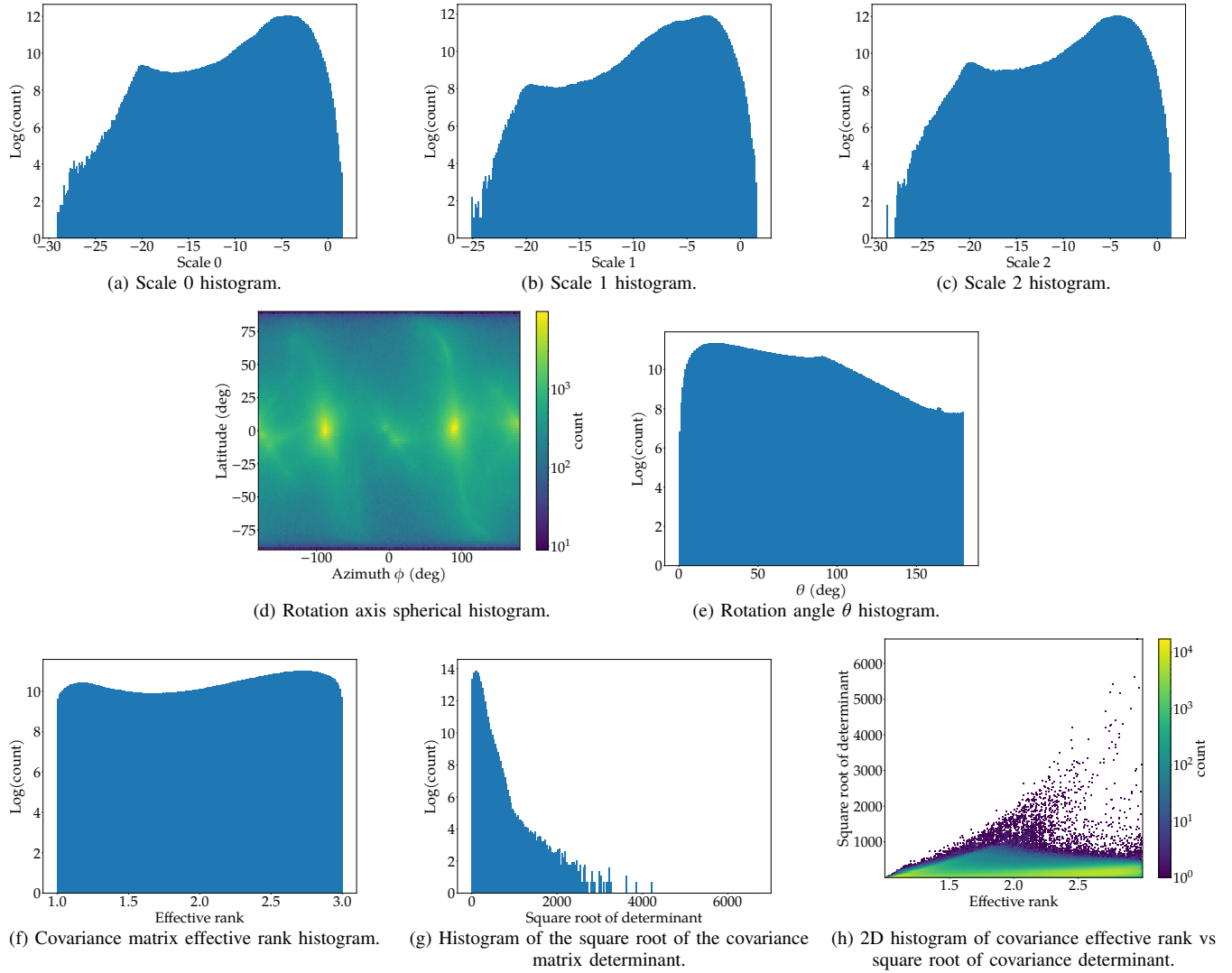


Fig. 12: Analysis of all static Gaussian splat reconstructions.

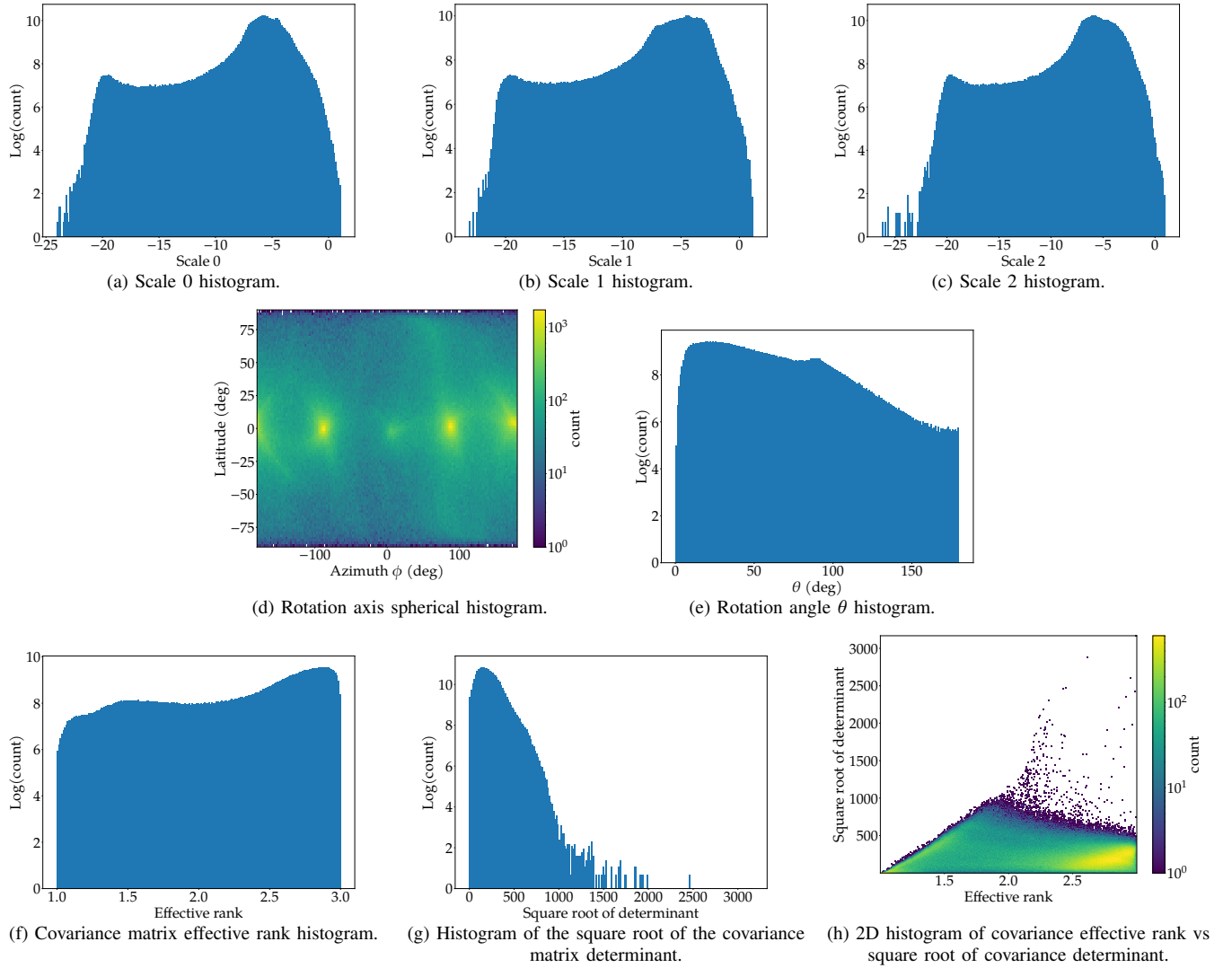


Fig. 13: Analysis of the Chapel static Gaussian splat reconstruction.

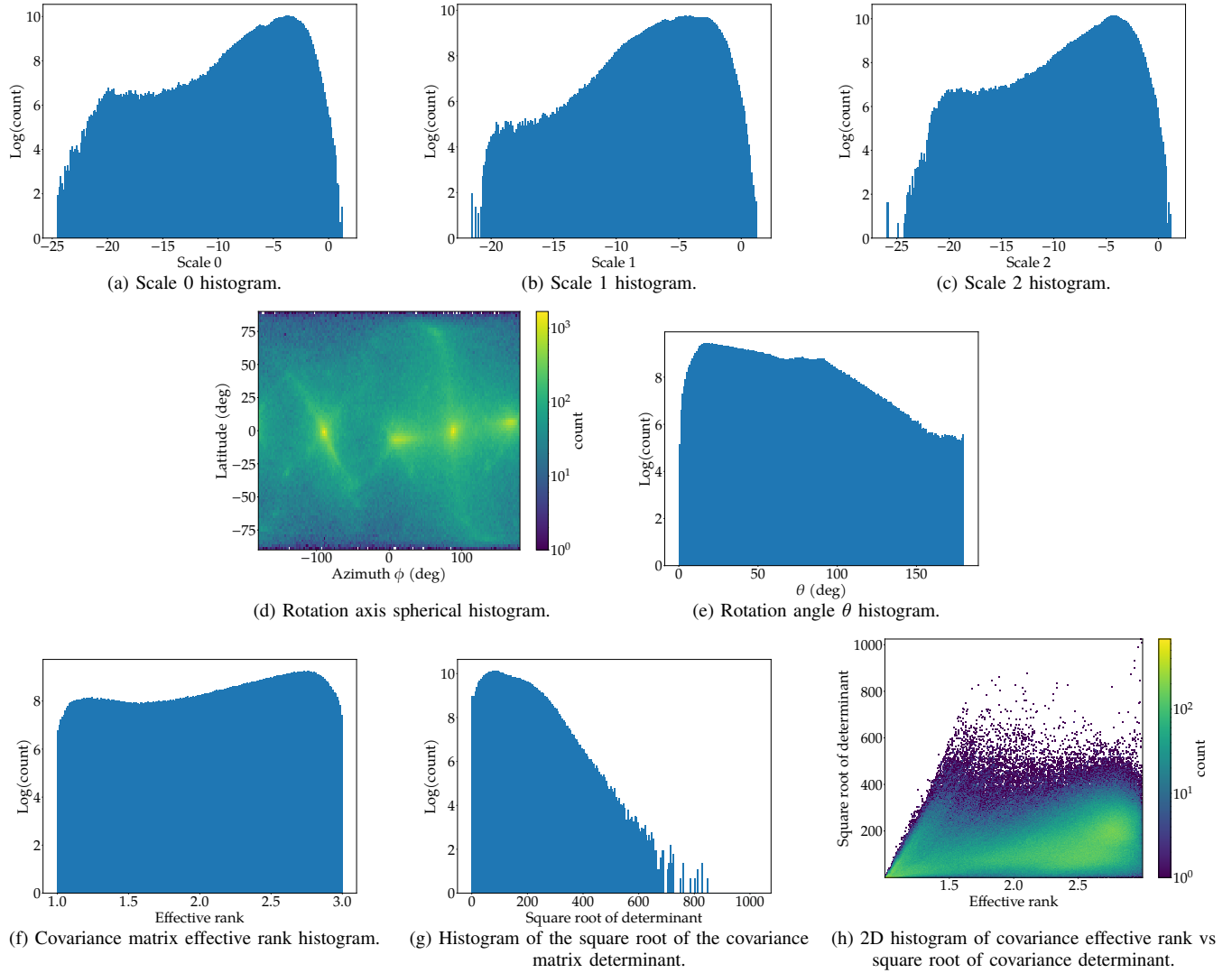


Fig. 14: Analysis of the Lecture Room static Gaussian splat reconstruction.

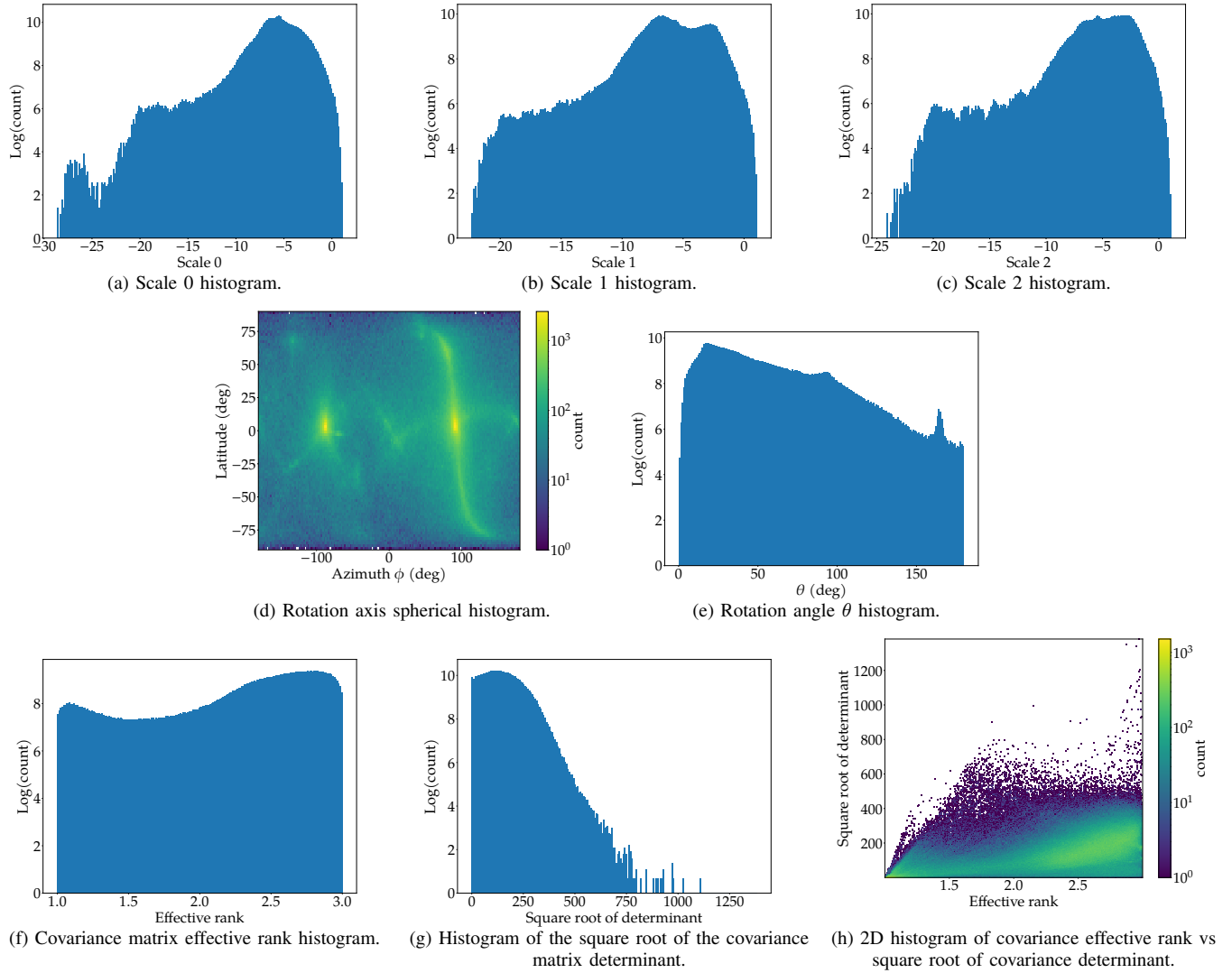


Fig. 15: Analysis of the Lecture Hall static Gaussian splat reconstruction.

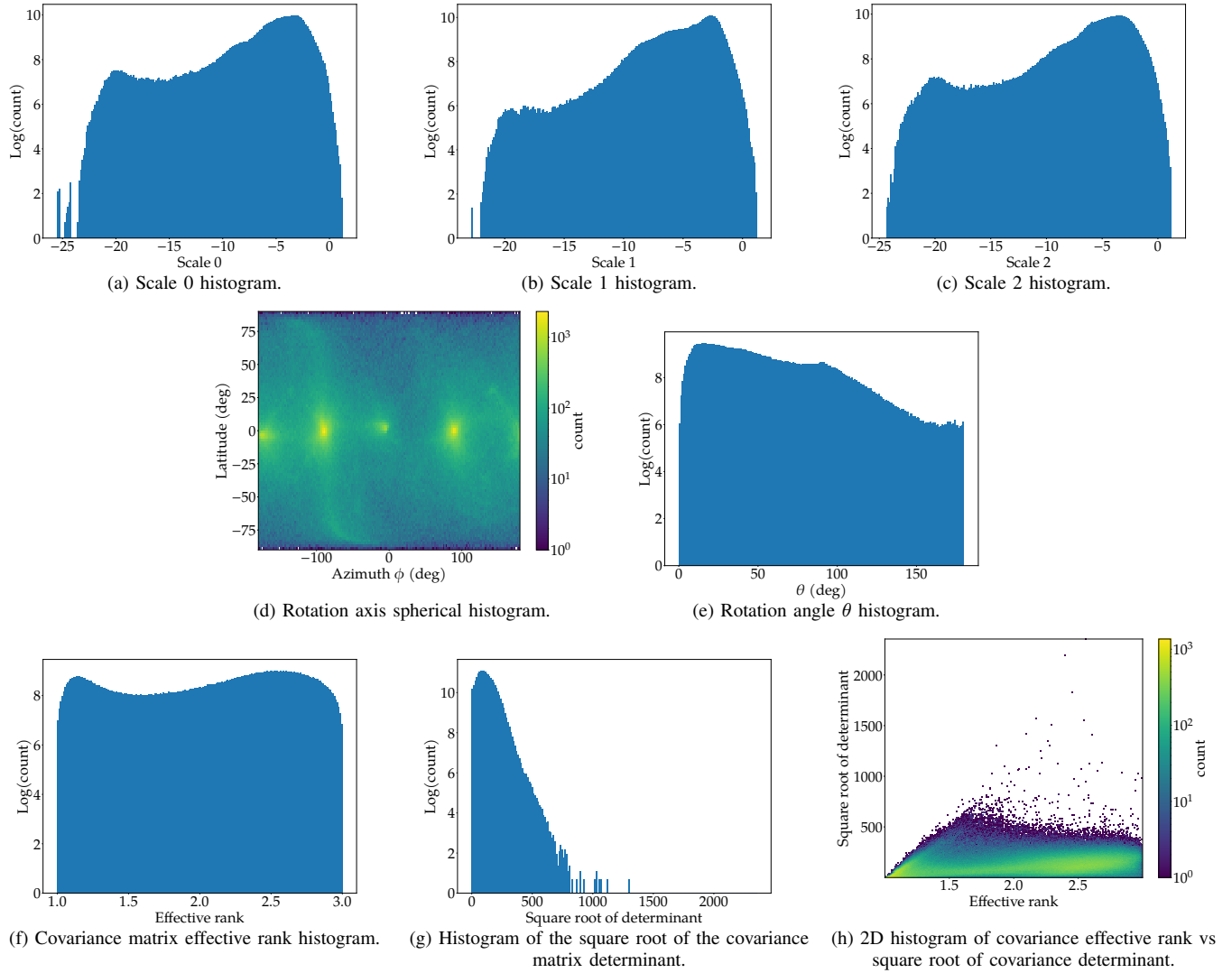


Fig. 16: Analysis of the Meeting Room static Gaussian splat reconstruction.

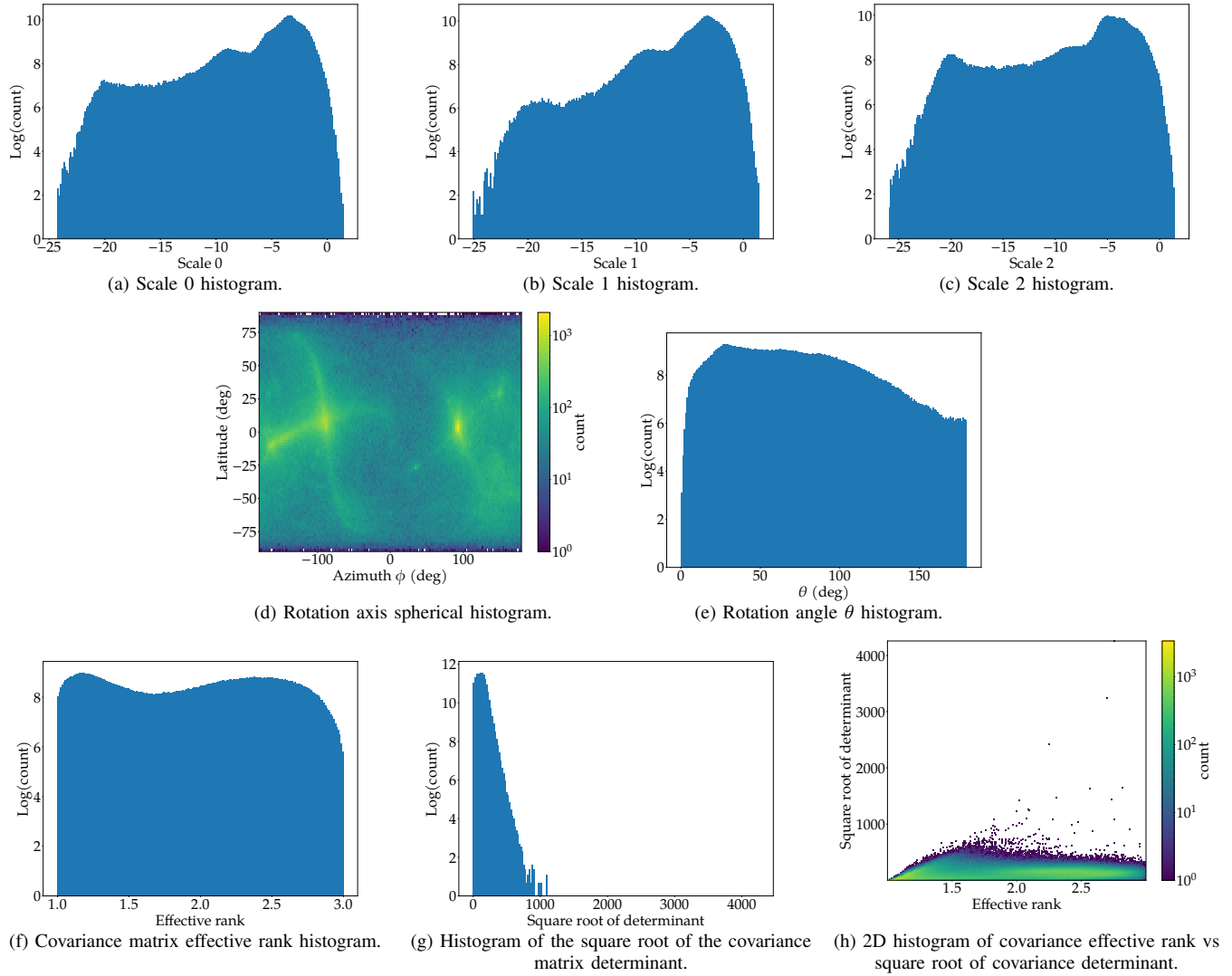


Fig. 17: Analysis of the Office Space static Gaussian splat reconstruction.

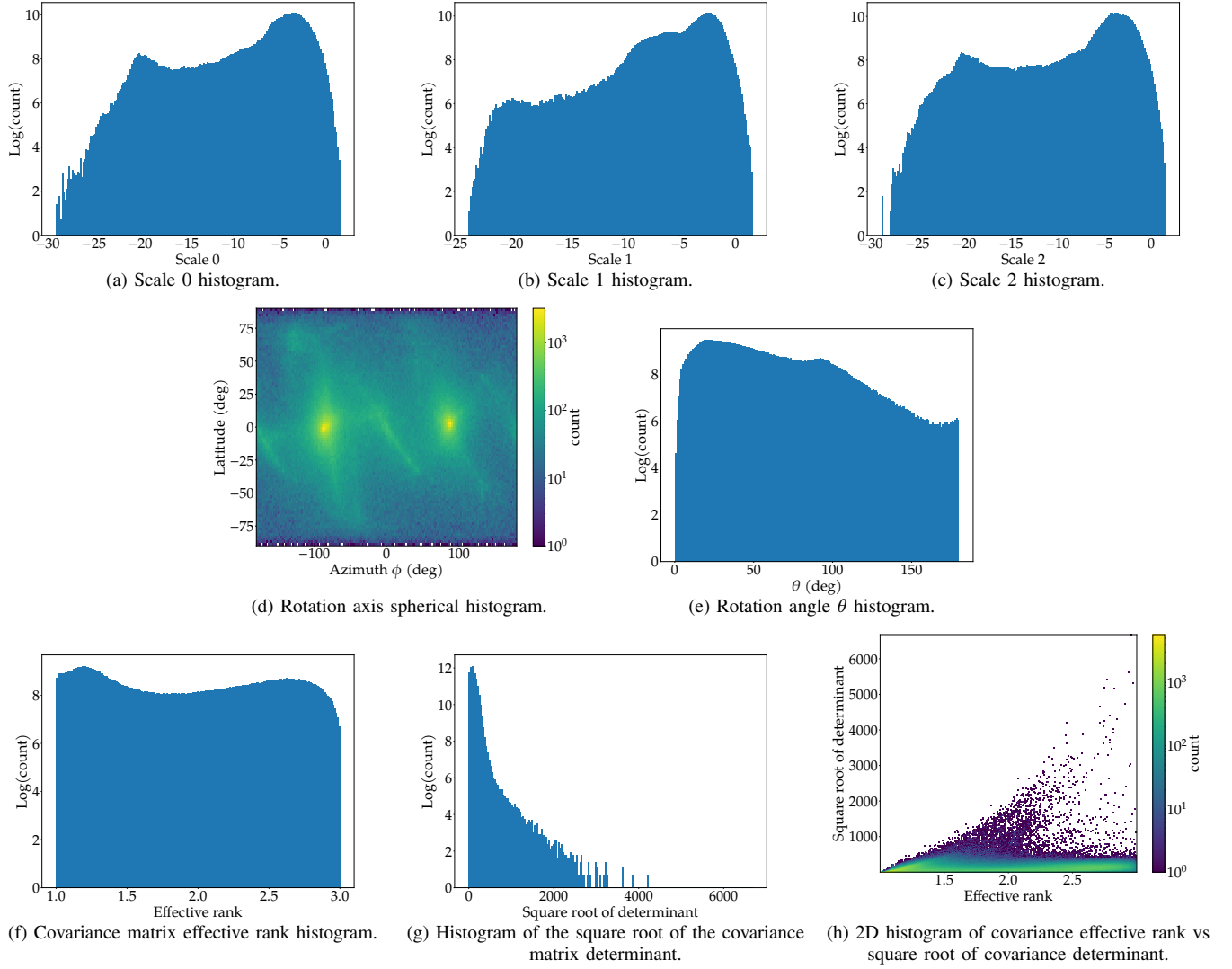
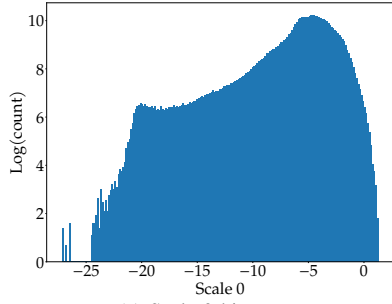
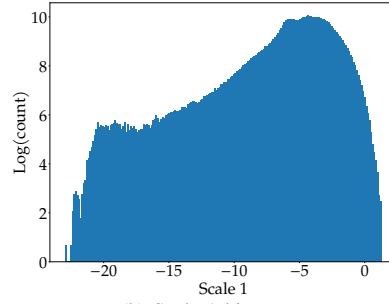


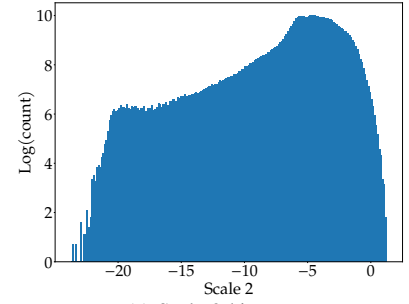
Fig. 18: Analysis of the Outdoor Scene static Gaussian splat reconstruction.



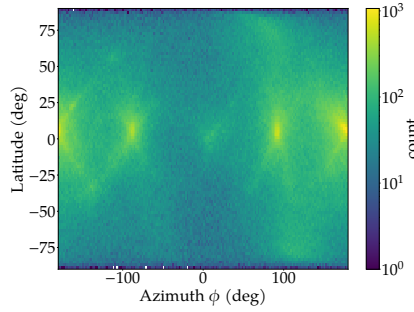
(a) Scale 0 histogram.



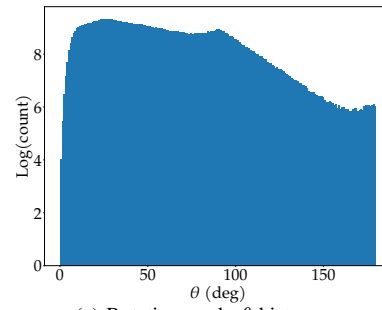
(b) Scale 1 histogram.



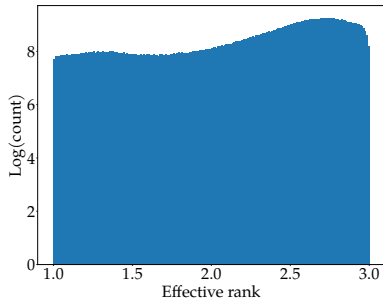
(c) Scale 2 histogram.



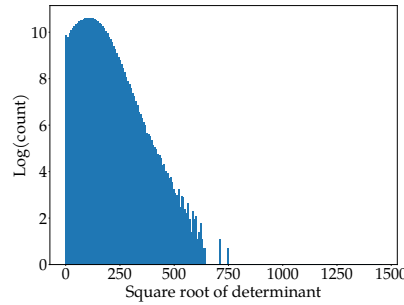
(d) Rotation axis spherical histogram.



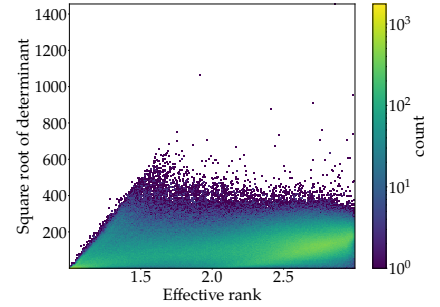
(e) Rotation angle θ histogram.



(f) Covariance matrix effective rank histogram.



(g) Histogram of the square root of the covariance matrix determinant.



(h) 2D histogram of covariance effective rank vs square root of covariance determinant.

Fig. 19: Analysis of the Cluttered Space static Gaussian splat reconstruction.